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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC)-MG1MDCFTQ100-FOOD SPOILAGE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap) Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions, Answer any 5 Questions, Each Question carries 1 mark

1. Comment on "Fusarium rot" in fruits and vegetables [U] / [CO5]

Fusarium rot is a common post-harvest fungal disease caused by *Fusarium* species. It usually enters through wounds or damaged tissues, leading to soft, watery decay with white to pinkish fungal growth. Affected fruits and vegetables rapidly deteriorate during storage due to tissue breakdown and mycotoxin production.

2. Yellow milk [K] / [CO5]

Yellow milk occurs due to the growth of pigment-producing bacteria such as *Pseudomonas syncyanea* or due to carotenoid pigments. These microorganisms break down milk components and produce yellowish discoloration, indicating spoilage and poor hygiene during handling or storage.

3. What is the significance of moisture content in food spoilage? [U] / [CO3]

Moisture content plays a crucial role in determining the susceptibility of food to spoilage. High moisture foods support rapid microbial growth, enzymatic activity, and chemical reactions that lead to deterioration. Low-moisture foods have longer shelf life because microorganisms cannot grow without sufficient water availability.

4. What is the economic impact of food spoilage? [K] / [CO1]

Food spoilage leads to direct financial losses due to discarded food, decreased product quality, and reduced shelf life. Industries incur additional costs for storage, handling, and spoilage control measures. Spoilage also affects consumer satisfaction and increases overall food wastage.

5. What conditions favour the growth of spoilage bacteria in poultry? [A] / [CO5]

Poultry meat is highly perishable due to its high moisture and nutrient content. Warm temperatures, inadequate chilling, poor hygiene, cross-contamination, and oxygen availability promote rapid bacterial growth. These conditions accelerate spoilage and reduce shelf life.

6. Enlist the common microorganisms that are associated with spoilage of meat. [U] / [CO5]

Meat is spoiled by a variety of microorganisms including *Pseudomonas*, *Clostridium*, *Lactobacillus*, *Shewanella putrefaciens*, and members of the Enterobacteriaceae family. These microbes break down proteins, fats, and carbohydrates, leading to off-odors, discoloration, and slime formation.

7. Which bacteria are commonly associated with fish spoilage? [An] / [CO5]

Spoilage of fish is mainly caused by psychrotrophic bacteria such as *Pseudomonas*, *Shewanella putrefaciens*, *Photobacterium phosphoreum*, and *Aeromonas* species. These organisms thrive at low temperatures and produce strong off-odors, slime, and discoloration.

[5x1=5]

Part B

Short Essay Type Questions, Answer any 4 Questions, Each Question carries 5 marks

8. What are the defects observed on fish spoilage? [U] / [CO5]

Fish spoils rapidly due to its high moisture, neutral pH, and abundant nutrients. Common spoilage defects include surface slime caused by bacterial growth, softening and breakdown of muscle tissues, and unpleasant odors such as fishy, sour, or ammonia-like smells. Visual changes include

dull eyes, pale gills, and discoloration of the skin or muscle. Advanced spoilage may cause belly bursting due to gas production from bacteria like *Shewanella*. Texture deterioration and loss of freshness are key indicators of spoilage.

9. What is the role of pH in food spoilage. How does it affect microbial growth enzymatic activity, and overall food safety? [U] / [CO3]

pH is a crucial intrinsic factor influencing food spoilage. Low pH (acidic foods) inhibits many spoilage bacteria and pathogens, allowing mainly yeasts and molds to grow. High pH foods, such as meat and fish, support rapid bacterial growth. pH also affects enzyme activity many enzymes function optimally around neutral pH, causing nutrient breakdown and chemical deterioration. Microbial toxins and spoilage metabolites vary with pH. By controlling pH through fermentation or acidification, food safety is enhanced and shelf life is prolonged.

10. What are mycotoxins, and how are they related to mold spoilage? [K] / [CO1]

Mycotoxins are toxic secondary metabolites produced by molds such as *Aspergillus*, *Penicillium*, and *Fusarium*. These toxins contaminate food during mold growth especially under warm, humid, or poorly stored conditions. Mycotoxins (like aflatoxins, ochratoxins, fumonisins) pose serious health risks such as carcinogenic, nephrotoxic, and immunosuppressive effects. Mold spoilage not only causes visible deterioration in foods such as grains, nuts, and fruits but also introduces these dangerous metabolites, making contaminated foods unsafe even if the mold is removed.

11. Comment on gas and alkali production in milk during spoilage. [A] / [CO5]

Milk spoilage involves many microorganisms that break down milk components. Gas production occurs mainly due to coliforms and *Clostridium* species, leading to frothiness, off-odors, and bloating of containers. Alkali production occurs due to proteolytic bacteria such as *Pseudomonas*, which degrade proteins into ammonia and amines. This increases the pH of milk, producing bitter flavors and causing curdling. Both gas and alkali formation are advanced stages of spoilage indicating microbial contamination and loss of milk quality.

12. Briefly explain the role of pests, insects, and rodents in spoilage. [U] / [CO4]

Pests, insects, and rodents significantly contribute to food spoilage by physically damaging food products, creating entry points for microbes, and directly contaminating food with their droppings, hair, and pathogens. They act as vectors carrying bacteria, fungi, and viruses from unclean environments to food storage areas. Their presence also increases local moisture and temperature, promoting microbial growth. Such contamination leads to reduced quality, health hazards, and economic losses in households and food industries.

13. Understanding food spoilage is crucial for ensuring food safety, maintaining nutritional quality, and minimizing economic losses. Justify. [K] / [CO1]

A clear understanding of food spoilage helps prevent foodborne illness by identifying early signs of microbial growth. It ensures that food retains its nutritional value and taste throughout its intended shelf life. Knowledge of spoilage mechanisms helps in designing proper storage, packaging, and processing techniques to reduce waste and financial losses. Understanding spoilage also aids in developing hygiene practices and quality control measures essential for consumer protection and industry standards.

14. Write a note on Ham and its spoilage. [K] / [CO5]

Ham, a cured meat product, is susceptible to both microbial and chemical spoilage. Despite curing, spoilage bacteria such as *Lactobacillus*, *Leuconostoc*, and *Pseudomonas* can grow on ham surfaces, causing slime formation, sour odors, and green discoloration. Chemical spoilage includes lipid oxidation leading to rancidity. Poor storage conditions, inadequate refrigeration, and contamination during slicing increase the spoilage rate. Spoiled ham shows off-odors, discoloration, tacky surfaces, and loss of characteristic flavor.

[4x5=20]

Part C

Long Essay Type Questions, Answer any 1 Question, Each Question carries 10 marks

15. Explain the role of bacteria, mold and yeast in food spoilage. [K] / [CO3]

Food spoilage is primarily caused by the activities of bacteria, molds, and yeasts, each contributing differently based on food type and storage conditions.

Bacteria:

Bacteria are responsible for rapid spoilage in protein-rich and moist foods such as meat, fish, milk, and vegetables. Psychrotrophic bacteria like *Pseudomonas* cause slime formation, surface discoloration, and off-odors in refrigerated foods. Proteolytic bacteria break down proteins into ammonia and sulfur compounds, causing putrefaction. Fermentative bacteria like lactic acid bacteria cause souring in milk and vegetables. Anaerobic bacteria like *Clostridium* cause gas production, bloating, and strong odors.

Molds:

Molds spoil foods with low moisture or acidic environments, including bread, fruits, and stored grains. They produce visible mycelial growth and pigments that discolor food. Molds cause spoilage through enzymatic breakdown of carbohydrates and organic acids. More importantly, many molds produce mycotoxins such as aflatoxins and ochratoxins, which pose serious health risks. Mold spoilage is common in improperly stored foods exposed to high humidity.

Yeasts:

Yeasts spoil high-sugar and acidic foods such as juices, syrups, jams, and pickles. They cause fermentation, gas production, and development of alcohols and esters that change taste and smell. Yeasts such as *Saccharomyces*, *Candida*, and *Zygosaccharomyces* thrive in low pH environments. Swelling of packaged food containers due to gas production is a common sign of yeast spoilage.

Together, these microorganisms degrade food quality, reduce shelf life, and pose potential health hazards.

16. Write a note on factors that influence the invasion of micro-organisms in meat. [An] / [CO4]

Microbial invasion in meat is affected by multiple intrinsic and extrinsic factors:

- **Animal factors:** The health, stress level, and hygiene of the animal before slaughter affect initial contamination. Diseased or stressed animals carry higher microbial loads.
- **Slaughterhouse and processing hygiene:** Contamination can occur from unclean equipment, surfaces, water, or handlers. Contact with hides, digestive contents, or soil introduces spoilage organisms.
- **Environmental factors:** Temperature, humidity, and air quality greatly influence microbial growth. Warm and humid environments accelerate spoilage.
- **Intrinsic factors of meat:** Meat has high moisture, neutral pH, and abundant nutrients that support microbial growth. The water activity (aw) of meat is favourable for bacteria and fungi.
- **Handling and storage:** Improper cooling, delayed refrigeration, cross-contamination, and poor packaging increase microbial invasion. Vacuum or modified-atmosphere packaging slows down spoilage but cannot prevent it completely.
- **Time-temperature abuse:** The rate of microbial multiplication increases rapidly if meat is stored above recommended temperatures.

Overall, spoilage can be minimized through strict hygiene, rapid chilling, and proper storage and packaging.

17. Evaluate the impact of microbial contamination on the shelf life of eggs? [E]/ [CO5]

Eggs, though protected by a shell, are highly vulnerable to microbial contamination when the shell integrity is compromised or hygiene is poor. Contaminating microorganisms such as *Pseudomonas*, *Proteus*, *Salmonella*, molds, and yeasts cause various types of spoilage including black rot, green rot, red rot, and fungal growth. These microorganisms degrade egg albumen, causing it to become watery, discolored, and foul-smelling. The yolk may also liquefy and mix with the albumen due to enzyme activity. Microbial contamination significantly reduces the shelf life of eggs as physical, chemical, and sensory qualities deteriorate quickly. Contaminated eggs also pose a serious health risk, especially from

Salmonella infections. Proper washing, sanitizing, cooling, and storage at low temperatures are essential to maintain egg quality and prevent spoilage.

[1x10=10]